

MORE DATA PRACTICE

SOLUTIONS

1. [3 marks]

Classify each numerical variable as a DISCRETE variable or a CONTINUOUS variable.

a) The numbers of brothers and sisters a person has.

D

b) The time it takes to travel from A to B in minutes.

C

c) A person's weight in kilograms.

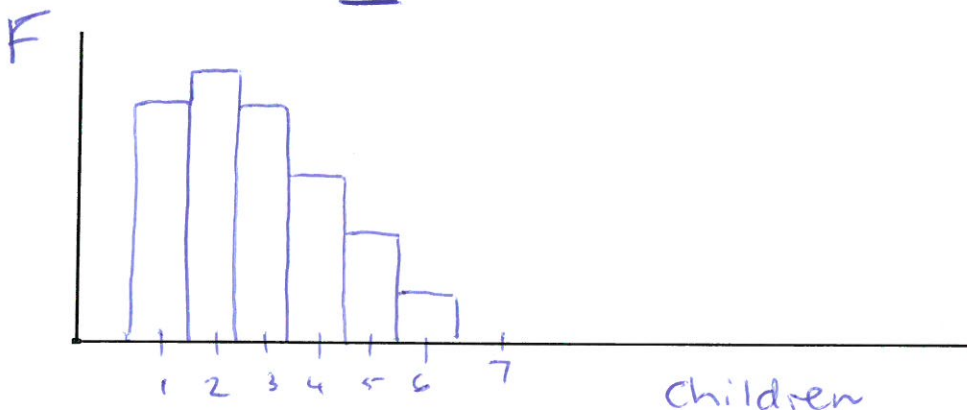
C

2. [4 marks]

Sketch what you consider to be a reasonable HISTOGRAM for the following:

The number of children in approximately 50 families, each of which have at least one child.

Your HISTOGRAM does need a scale on the Horizontal axis.



3. [4, 6 marks]

Thirteen students completed a "Standing Jump Test" and the distances (in cm) they jumped are recorded in the Stem and Leaf plot below.

Calculate the Mean, Mode, Median and Range for the data.

23	1 3
24	4 4 4
25	2
26	5
27	7 7 8
28	6 9
29	0

$$\text{Range} = 290 - 231 \\ = 59 \text{ cm}$$

$$\text{Mode} = 244 \text{ cm}$$

$$\text{Median} = 7^{\text{th}} \text{ score } 265 \text{ cm}$$

$$\text{Mean} = \frac{3410}{13} = 262.3 \text{ cm}$$

3. b) Construct a MEDIAN BOX and WHISKER plot for the same data,

4. [4 marks]

These results are for Duncraig SHS for the Semester One Year 9 Examination in 2015.

Class	Mean	Number of students
Class A	62.45	25
Class B	47.17	28
Class C	58.73	30

Use the above results to calculate the overall examination average for these three classes. You must show your working to get full marks. Give your answer accurate to 1 decimal place.

$$\begin{aligned}\bar{x} &= \frac{(62.45 \times 25) + (47.17 \times 28) + (58.73 \times 30)}{25 + 28 + 30} \\ &= \frac{4643.91}{83} \\ &= 56.0 \text{ (10 p)}\end{aligned}$$

5. [2, 1, 1, 3 marks]

The following are 8 prices (to the nearest \$1 000) for homes sold in Duncraig over the last year.

\$567 000 \$523 000 \$534 000 \$712 000
\$576 000 \$599 000 \$723 000 \$596 000

- a) Find
- the Mean Price (to the nearest \$100). $\$603\,800$
 - the Median Price. $\$586\,000$

b) How many house prices are below the Mean Price? 6

c) How many house prices are below the Median Price? 4

d) A ninth house is sold for \$650 000. Which measurement, Mean or Median, changes the most? You must have suitable reasoning or results to support your statement.

$$\begin{aligned}\text{New } \bar{x} &= 608\,889 \\ \text{med} &= 596\,000\end{aligned}$$

$\therefore$ Median changed the most

6. [3 marks]

Using the given data for the two distributions (which use Raw Scores not percentages), decide which Student Score is the better score and explain why?

Distribution A	Raw Mean = 40	Standard Deviation = 6	Student score = 46
Distribution B	Raw Mean = 64	Standard Deviation = 8	Student score = 70

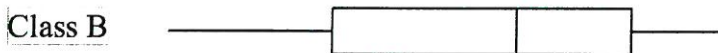
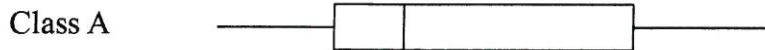
$$A) \frac{46 - 40}{6} = 1$$

$$B) \frac{70 - 64}{8} = 0.75$$

∴ A is a better score as it is more standard deviations above the mean, i.e. has a higher standard score.

7. [6 marks]

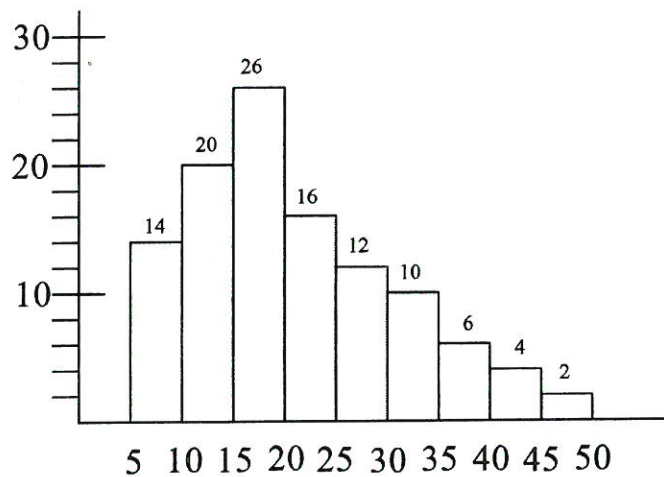
Compare the two MEDIAN BOX PLOTS shown below.



- Median of B is 57, higher than that of A with 52
- Both classes have a range of 24
- Q1 is 49 for both classes
- Q3 is 62 for both classes
- class B is negatively skewed
- class A is positively skewed.
- Both classes have the same IQR of 13.

8. [4 marks]

Describe the distribution shown in the Histogram.



- 110 Data points in total.
- Unimodal, modal class of 15-20
- Range of 45.
- Positively skewed
- Mean is larger than median due to the skewness.

9. [3 marks]

For the data set below calculate the Mean and Standard Deviation, accurate to 1 decimal place. (These are only estimates of the actual values).

Interval	Number of students
$20 \leq x < 30$	3
$30 \leq x < 40$	7
$40 \leq x < 50$	12
$50 \leq x < 60$	10
$60 \leq x < 70$	5

$$\text{Mean} = 46.9 \text{ (1dp)}$$

$$\text{s.d } \sigma_x = 11.4 \text{ (1dp)}$$